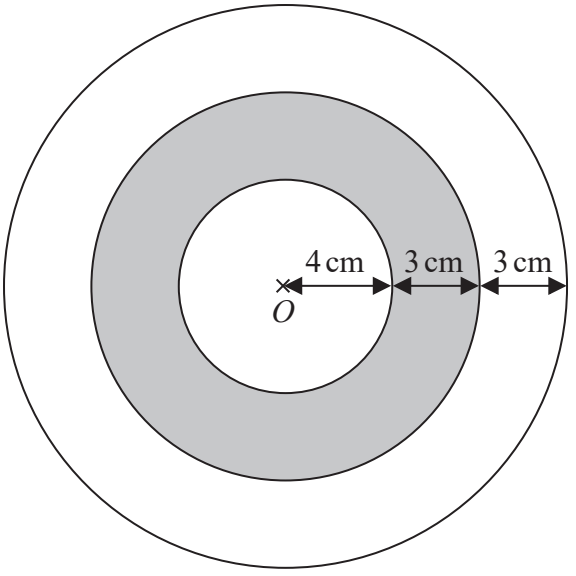
4 The diagram shows a logo made from three circles.



Each circle has centre O.

Daisy says that exactly $\frac{1}{3}$ of the logo is shaded.

Is Daisy correct?
You must show all your working.

The three radii are 4, 4 + 3 = 7 and 7 + 3 = 10 cm.

Middle area = $\pi \times 7^2 = 49\pi$ cm^2.

1 mark
Use πr^2

Inner area = $\pi \times 4^2 = 16\pi$ cm^2.

Shaded area = $49\pi - 16\pi = 33\pi$ cm^2.

1 mark
Subtract the inner area

Shaded fraction = $\frac{33\pi}{100\pi} = \frac{33}{100}$

1 mark
Compare with whole area 100π

No. 33/100 = 0.33, whereas 1/3 = 0.333...

1 mark
The shaded fraction is smaller than exactly one third

(Total for Question 4 is 4 marks)